

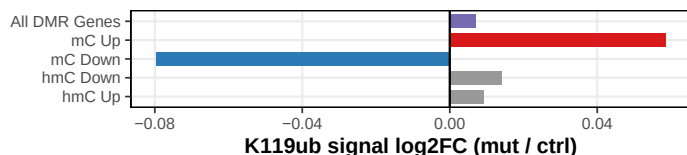
Figure 7. Quantitative cascade: BAP1 loss -> H2AK119ub -> TET-block -> methylation -> MeCP2

Each panel quantifies one mechanistic link (deep-seq, 8 samples, sex covariate)

A

BAP1 loss -> H2AK119ub

Global K119ub rise: background median log2FC = +0.007 ($p = 1.81 \times 10^{-20}$)

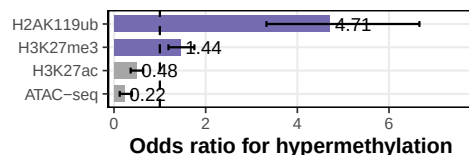


Background (all DMR genes) hmC genes mC hyper genes mC hypo genes

B

H2AK119ub -> hypermethylation

Differential-mark logistic model: H2AK119ub OR = 4.71

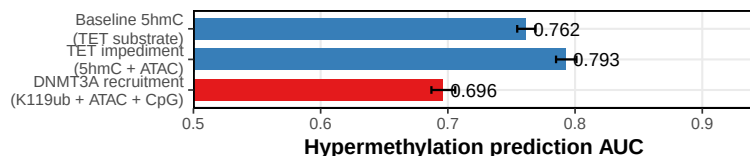


Promotes hyper (OR > 1) Suppresses hyper (OR < 1)

C

TET impediment, not DNMT3A recruitment

TET AUC = 0.793 vs DNMT3A AUC = 0.696 (DeLong $p = 9.43 \times 10^{-9}$)

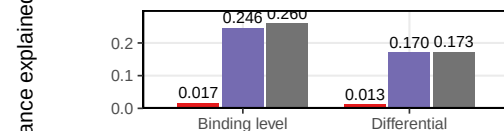


DNMT3A pathway TET pathway

D

DNA methylation -> MeCP2 binding

MeCP2 binds chromatin: chromatin $R^2 = 0.246$ vs CG $R^2 = 0.017$

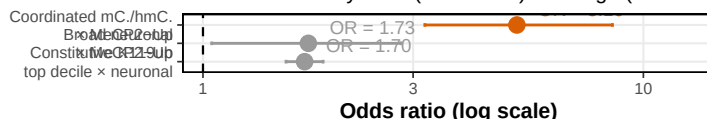


CG only Chromatin only Full (CG + Chromatin)

E

Convergence on neuronal genes

MeCP2 tracks methylation (OR = 5.16) > lineage (OR = 1.73)



Lineage Methylation-driven